



# Non-asymptotic confidence bands on the probability an individual benefits from treatment

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## Introduction

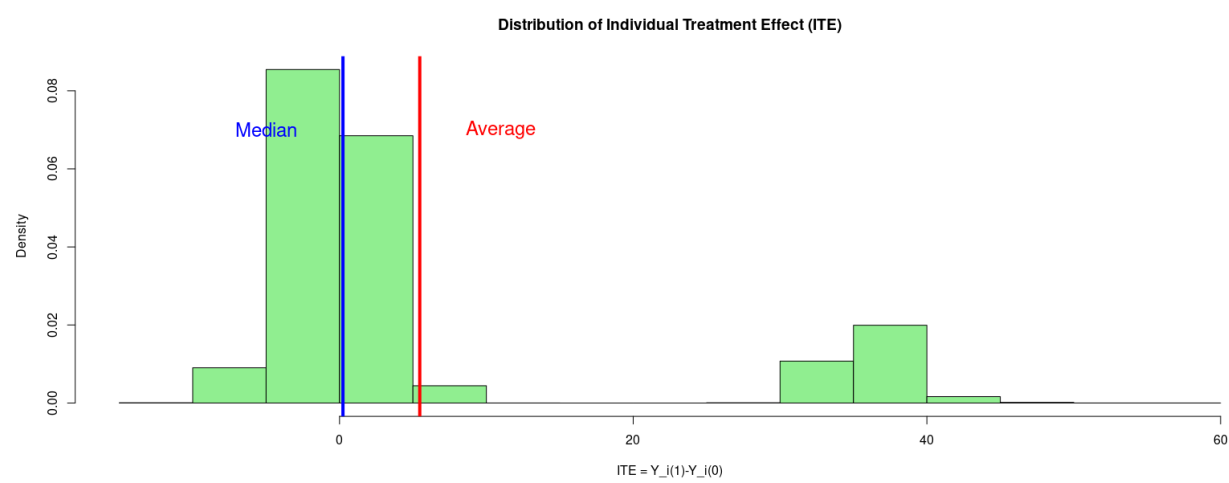


Figure 1: Hypothetical distribution of  $Y_i(1) - Y_i(0)$ .

In [1], we seek **closed-form**, non-asymptotic confidence bands on the **inestimable** probability an individual benefits from treatment (**PIBT**).

## Target Bounds to Estimate

1.  $\theta_{\delta}^L \leq \Pr(Y_i(1) - Y_i(0) > \delta) \leq \theta_{\delta}^U$ .
2.  $\theta_{\delta,x}^L \leq \Pr(Y_i(1) - Y_i(0) > \delta | X_i = x) \leq \theta_{\delta,x}^U$ .

If more appropriate, may instead bound

$$\Pr(Y_i(1)/Y_i(0) > \delta) \text{ or } \Pr(Y_i(1) = 1, Y_i(0) = 0).$$

## Insight 1: Practical Interpretation

With  $(1 - \beta) \times 100\%$  confidence, PIBT is contained between

[Estimator for lower bound on PIBT]  
– [Margin of error for PIBT bounds]

and

[Estimator for upper bound on PIBT]  
+ [Margin of error for PIBT bounds].

$$\geq \sup_{\delta} \{ |\hat{\theta}_{\delta}^L - \theta_{\delta}^L| \vee |\hat{\theta}_{\delta}^U - \theta_{\delta}^U| \}$$

## Insight 2: Power Analysis For RCT

- Margin of error for PIBT

bounds =  $\left(\frac{\log(4/\alpha)}{2}\right)^{\frac{1}{2}} \left(n_0^{-\frac{1}{2}} + n_1^{-\frac{1}{2}}\right)$  with  $(1 - \alpha) \times 100\%$  confidence.

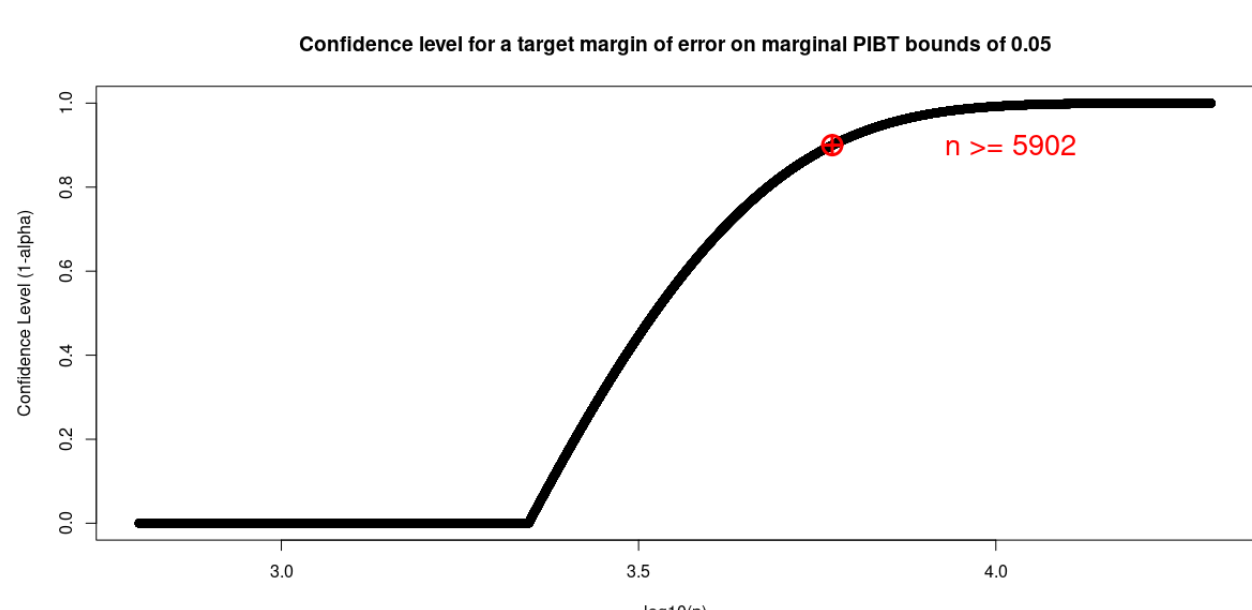


Figure 2: Power analysis based on our Theorem 1 in [1].

## Insight 3: Heterogeneity in PIBT

- Simultaneous Inference at all strata  $x$ .

- Estimate bounds via **imputation**:

$$\hat{\mu}_w(x) + \{Y_i(w) - \hat{\mu}_w(X_i)\},$$

an approximate draw from distribution of  $Y_i(w) | X_i = x$ .

- Confidence level correction:

$$1 - \alpha - \Pr\left(\bigvee_{w=0,1} \sup_x |\hat{\mu}_w(x) - \mu_w(x)| > t_w\right)$$

for uniform PIBT coverage.

At sample size  $n$ , what is the uniform Margin of Error on efficacy bounds at the 90% confidence level?

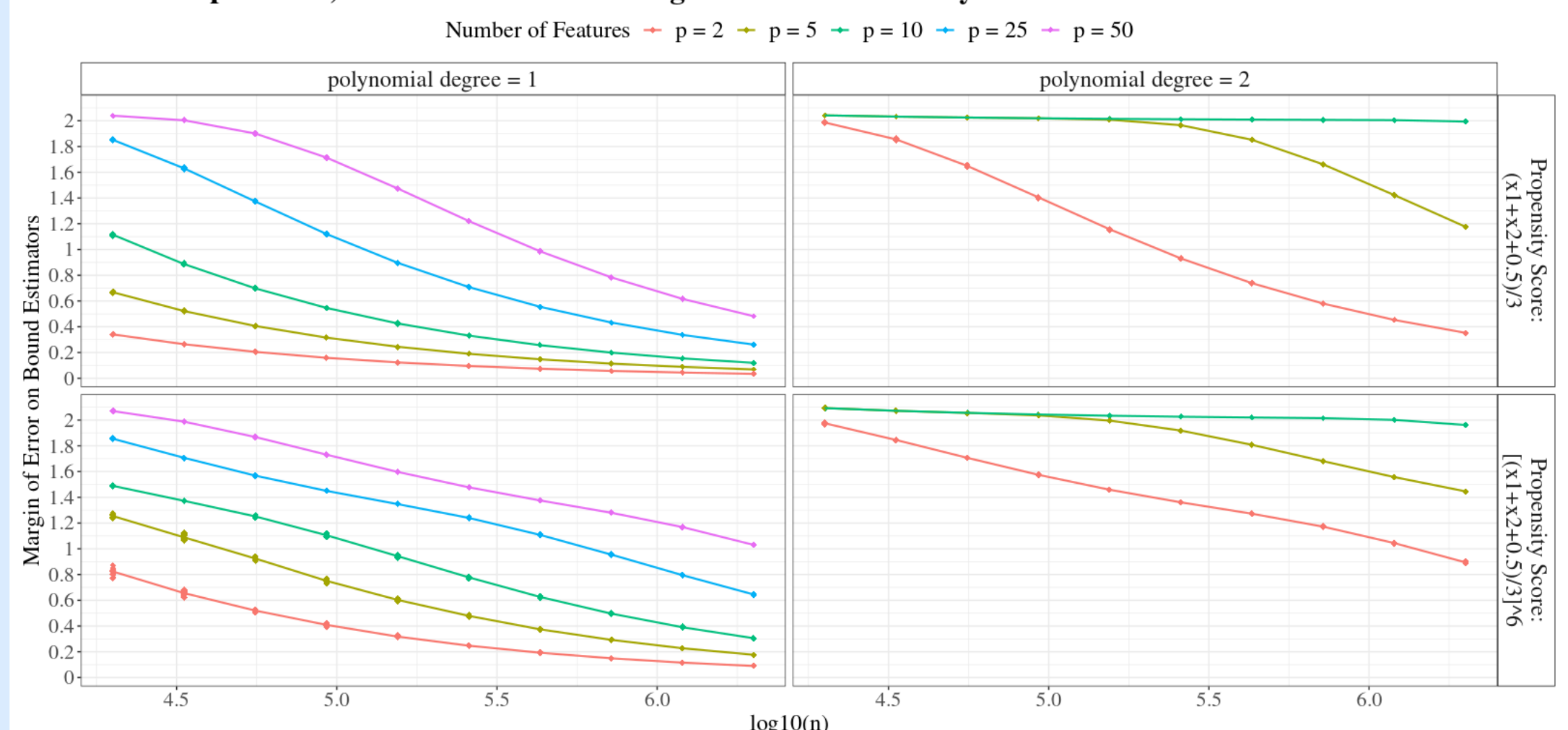


Figure 3: Power analysis based on our Theorem 3 in [1] applied to canonical linear model.

## Methodology

- Extension of existing **plug-in estimation** framework [2].
- **Distribution-free** bounds on CDF of difference [3] (any possible association).
- Non-parametric Empirical CDF deviation inequality [4].
- Neyman-Rubin causal model; marginal and conditional ignorability.

## References

- [1] Ruiz, G and Padilla, OHM (2022). “Non-asymptotic confidence bands on the probability an individual benefits from treatment (PIBT).” *arXiv:2205.09094*.
- [2] Fan, Y and Soo Park, S (2010). “Sharp Bounds on the Distribution of the Treatment Effect and Their Statistical Inference.” *Econometric Theory*.
- [3] Williamson, RD and Downs, T (1990). “Probabilistic arithmetic. I. Numerical methods for calculating convolutions and dependency bounds.” *International Journal of Approximate Reasoning*.
- [4] Massart, P (1990). “The Tight Constant in the Dvoretzky-Kiefer-Wolfowitz Inequality.” *The Annals of Probability*.